THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(c)** 2310	Highest powers of $2 \times 3 \times 5$, $2 \times 3 \times 7$, $2 \times 3 \times 11$
2	**(a)** 28	$(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)$
3	**(b)** -4	$k^2 = 4p^2$; $k = 2p$ gives coincident lines, $k = -2p$ gives parallel lines
4	**(b)** 29	$(\alpha+\beta)^2 - 2\alpha\beta$
5	**(d)** 0	Common difference is -1; $a_{p+q} = 0$
6	**(d)** 8	$(m+n)^2 - m^2$
7	**(a)** 12	Let the point be $(0, y)$
8	**(b)** $\frac{25}{9}$	$1 + \tan^2\theta$
9	**(c)** $2+2\sqrt{3}$ m	$h(\cot 30^\circ - \cot 45^\circ) = 4$ gives $h = 4 \div (\sqrt{3}-1)$; Height $= 2+2\sqrt{3}$ m (distances $6+2\sqrt{3}$ m and $2+2\sqrt{3}$ m)
10	**(d)** 48 cm	$AM = \frac{30\times40}{50} = 24$ cm; $AB = 2AM = 48$ cm
11	**(c)** 4.47 cm²	Sector $= 25.67$ cm²; triangle $= \frac{\sqrt{3}}{4}r^2 = 21.19$ cm²; Segment $= 25.67 - 21.19 = 4.47$ cm²
12	**(a)** 4410 cm³	Sphere $= \frac{4}{3}\pi\left(\frac{21}{2}\right)^3 = 4851$ cm³; cube $= 9261$ cm³; Wasted $= 4410$ cm³
13	**(d)** 23	$\frac{4325+25p}{77+p} = 49$ gives $(25-49)p = -552$; $p = 23$
14	**(d)** $\frac{5}{36}$	Favourable outcomes: (2,6), (3,5), (4,4), (5,3), (6,2); number $= 5$; $P = \frac{5}{36}$
15	**(b)** 60	LCM of 1 to 6
16	**(b)** 30	$\alpha\beta(\alpha+\beta)$
17	**(a)** Both A and R are true and R is the correct explanation of A.	HCF x LCM = 6 x 36 = 216 = 12 x 18, and R is the general result used to find the LCM.
18	**(a)** Both A and R are true and R is the correct explanation of A.	Each point where the graph meets the x-axis has $y = 0$, i.e. $p(x) = 0$; so R explains A.
19	**(a)** Both A and R are true and R is the correct explanation of A.	$3+1=4$ and $3-1=2$; the common point satisfies both equations, which is exactly the solution (R).
20	**(a)** Both A and R are true and R is the correct explanation of A.	$D = 9 - 40 = -31 < 0$, so there are no real roots, as stated in R.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	Numbers are $12 \times 3 = 36$ and $12 \times 5 = 60$ (1); LCM $= 180$ (1).	1+1
22	New sum $= \alpha^2 + \beta^2 = 10$ (1); product $= (\alpha\beta)^2 = 9$; polynomial $x^2 - 10x + 9$ (1).	1+1
23	$\frac{k}{8} = \frac{2}{k} \Rightarrow k = \pm 4$ (1); $k = 4$ gives coincident lines, so $k = -4$ (1).	1+1
24	$D = (4)^2 - 4k(1) = 0$ (1); $k = 4$ (1).	1+1
25	$a + 3d = 9$, $a + 8d = 4$ (1); $d = -1$, $a = 12$ so $a_{13} = 0$ (1).	1+1
26	$\triangle ADE \sim \triangle ABC$ (1); $\frac{DE}{BC} = \frac{AD}{AB} = \frac{3}{7}$ (1); $DE = 6$ cm (1).	1+1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	LCM$(18, 24, 36) = 72$ min (2); rounds 4, 3, 2 (1).	2+1
28	$\alpha+\beta = 3$, $\alpha\beta = 2$ (1); sum $= (\alpha+\beta) + \frac{\alpha+\beta}{\alpha\beta} = \frac{9}{2}$ (1); product $= \alpha\beta + 2 + \frac{1}{\alpha\beta} = \frac{9}{2}$ (1).	1+1+1
29	Put $p = \frac{1}{x}$, $q = \frac{1}{y}$: $2p + 3q = 2$, $3p + 5q = \frac{19}{6}$ (1); eliminating q: $(1)p = \frac{1}{2}$ so $p = \frac{1}{2}$ (1); $q = \frac{1}{3}$ (1); $x = 2$, $y = 3$ (1).	1+1+1+1
30	LHS $= \frac{2(x^2+1)}{x^2-1}$ (1); cross-multiplying: $3(x^2+1) = 5(x^2-1)$ (1); $x^2 = 4$ so $x = \pm 2$ (1).	1+1+1

31	$\frac{6}{2}(2a+5d)=75$ and $\frac{10}{2}(2a+9d)=185$ (1); simplify to linear equations in a, d (1); $a=5$, $d=3$, AP $5, 8, 11, 14, \ldots$ (1).	1+1+1
	OR: $\triangle AOB \sim \triangle COD$ (AA) (1); $\dfrac{AO}{OC}=\dfrac{AB}{DC}=2:3$ (1); area ratio $=4:9$ (1).	1+1+1
32	$AB^2=25$, $BC^2=25$, $CD^2=25$, $DA^2=25$ (2); all four sides equal (1); diagonals $AC^2=90$, $BD^2=10$ differ so it is a rhombus, not a square (2).	2+1+2
	OR: LHS $=\dfrac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta}$ (2); $=\dfrac{1}{\sin\theta\cos\theta}=\sec\theta\csc\theta$ (1).	2+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $144=2^4\times3^2$, $180=2^2\times3^2\times5$ (1). (ii) HCF = $2^2\times3^2=36$ chairs per row (1). (iii) Rows: $144/36+180/36=4+5=9$ (1). (iv) $216=2^3\times3^3$; HCF(144, 180, 216) = $2^2\times3^2=36$ (1); rows $=4+5+6=15$ (1).	1+1+1+2
34	(i) $x^2-8x+15=(x-3)(x-5)$, so zeros are 3 and 5 (1). (ii) Sum $=8$, product $=15$ (1). (iii) Twice, at $(3,0)$ and $(5,0)$ (1). (iv) Reciprocal zeros $\frac{1}{3}, \frac{1}{5}$: sum $=\frac{8}{15}$, product $=\frac{1}{15}$, polynomial $x^2-\frac{8}{15}x+\frac{1}{15}$, i.e. $15x^2-8x+1$ (2).	1+1+1+2
35	Let boat x km/h, stream y km/h; put $p=\frac{1}{x-y}$, $q=\frac{1}{x+y}$: $36p+40q=10$, $24p+70q=11$ (2); $p=\frac{1}{6}$, $q=\frac{1}{10}$ (1); $x-y=6$, $x+y=10$ (1); boat 8 km/h, stream 2 km/h (1).	2+1+1+1
36	Smaller tap x h, larger $x-5$ h: $\dfrac{1}{x}+\dfrac{1}{x-5}=\dfrac{1}{\frac{10}{3}}$ (2); simplify to a quadratic and solve, rejecting the root less than 5: $x=10$ (2); times 10 h and 5 h (1).	2+2+1
	OR: AP with $a=100$, $l=40$, $n=21$ (1); $S_n=\frac{21}{2}(100+40)=1470$ cm (2); middle term $=\frac{100+40}{2}=70$ cm (2).	1+2+2
37	Right triangle with legs 60 and 80 (1); distance $=\sqrt{3600+6400}=100$ km (2); half-distance triangle is similar with ratio $1:2$, so the distances are in ratio $2:1$ (2).	1+2+2
	OR: Centroid $=(6, \frac{8}{3})$ (2); area $=\frac{1}{2}\lvert\ldots\rvert=48$ sq. units (3).	2+3
38	$\csc\theta-\sin\theta=\dfrac{\cos^2\theta}{\sin\theta}$, $\sec\theta-\cos\theta=\dfrac{\sin^2\theta}{\cos\theta}$ (2); $\tan\theta+\cot\theta=\dfrac{1}{\sin\theta\cos\theta}$ (1); product $=\sin\theta\cos\theta\cdot\dfrac{1}{\sin\theta\cos\theta}=1$ (2).	5
	OR: Let the tower be H and the distance d: $H=d\tan60^\circ$ and $H-4=d\tan45^\circ$ (2); Subtract: $d(\tan60^\circ-\tan45^\circ)=4$, so $d=2+2\sqrt{3}$ (2); $H=d\tan60^\circ=6+2\sqrt{3}$; tower $6+2\sqrt{3}$ m, distance $2+2\sqrt{3}$ m (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER

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ANSWER KEY & MARKING SCHEME SET 2

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** 63	Twice the area
2	**(c)** $-\frac{7}{25}$	$\cos^2\theta - \sin^2\theta$
3	**(a)** 20 m	Height of the top of the flagstaff $= d\tan 60^\circ = 30$ m; Length $= 30 - 10 = 20$ m
4	**(b)** 14 cm	Perimeter $= PC + CE + ED + DP = PC + CA + DB + DP = PA + PB$; Perimeter $= 2 \times 7 = 14$ cm
5	**(d)** 150 mm	Circumference $= \pi d = 66$ mm; diameters $= 4 \times 21 = 84$ mm; Total $= 150$ mm
6	**(b)** 2425.5 cm³	Cylinder $= \pi r^2(2r) = 7276.5$ cm^3; sphere $= \frac{4}{3}\pi r^3 = 4851$ cm^3; Unoccupied $= 2425.5$ cm^3
7	**(b)** 9	$238.4 = 230 + \frac{50-(20+x)}{25} \times 10$ gives $x = 9$; $x = 9$; then $y = 100 - 82 - x = 9$
8	**(b)** $\frac{1}{6}$	Favourable outcomes: (1,1), (1,2), (1,3), (2,1), (2,2), (3,1); number $= 6$; $P = \frac{6}{36} = \frac{1}{6}$
9	**(d)** 9	HCF of the differences from the remainder
10	**(c)** 7	$\sqrt{(\alpha+\beta)^2 - 4\alpha\beta}$
11	**(d)** 8	Solve for x and y
12	**(b)** 6	$D = 0$ gives $k^2 = 4m$
13	**(a)** 46	$a_n = S_n - S_{n-1}$
14	**(d)** $\frac{12}{5}$	$\frac{1}{2}ab = \frac{1}{2}ch$
15	**(c)** $\frac{16}{3}$	Equate squared distances
16	**(c)** $\frac{24}{25}$	$2 \cdot \frac{p}{c} \cdot \frac{q}{c}$
17	**(a)** Both A and R are true and R is the correct explanation of A.	$7 - 3 = 11 - 7 = 15 - 11 = 4$; R is the defining property of an AP.
18	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{2}{3} = \frac{4}{EC} \Rightarrow EC = 6$; R is the Basic Proportionality Theorem.
19	**(a)** Both A and R are true and R is the correct explanation of A.	$\sqrt{9+16} = 5$; R is the formula used.
20	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{3}{4} + \frac{1}{4} = 1$; A is a particular case of the identity in R.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$AB^2 = 25$, $BC^2 = 25$, $CD^2 = 25$, $DA^2 = 25$ (2); all four sides equal (1); diagonals $AC^2 = 80$, $BD^2 = 20$ differ so it is a rhombus, not a square (2).	2+1+2
22	$\cos^2\theta = 1 - \left(\frac{4}{5}\right)^2 = \frac{9}{25}$ (1); value $= -\frac{7}{25}$ (1).	1+1
23	Let height $= h$; the distances from the foot are $h\cot 45^\circ$ and $h\cot 60^\circ$, which differ by 6 m; $h(\cot 45^\circ - \cot 60^\circ) = 6$ gives $h = 6 \div (1 - \frac{\sqrt{3}}{3})$ (1); Height $= 9 + 3\sqrt{3}$ m (distances $9 + 3\sqrt{3}$ m and $3 + 3\sqrt{3}$ m) (1).	1+1
24	Area of $\triangle OAP = \frac{1}{2} \times 60 \times 45$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB); $AM = \frac{60 \times 45}{75} = 36$ cm (1); $AB = 2AM = 72$ cm (1).	1+1
25	The 14 ribs divide the circle into 14 equal sectors; Area of the circle $= \frac{22}{7} \times 49^2 = 7546$ cm^2 (1); Area between ribs $= 7546 \div 14 = 539$ cm^2 (1).	1+1
26	Volume of water displaced $=$ volume of sphere $=$ rise $\times$ base area; $\frac{4}{3}\pi(6)^3 = \pi(8)^2 x$ (1); $x = 4.5$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Let the tower be H and the distance d: $H = d\tan 60^\circ$ and $H - 18\sqrt{3} = d\tan 30^\circ$ (1); Subtract: $d(\tan 60^\circ - \tan 30^\circ) = 18\sqrt{3}$, so $d = 27$ (1); $H = d\tan 60^\circ = 27\sqrt{3}$; tower $27\sqrt{3}$ m, distance 27 m (1).	1+1+1

28	Area of $\triangle OAP = \frac{1}{2} \times 30 \times 40$ and also $= \frac{1}{2} \times OP \times AM$ where $AM \perp OP$ (M is the midpoint of AB) (1); $AM = \frac{30 \times 40}{50} = 24$ cm (1); $AB = 2AM = 48$ cm (1).	1+1+1
29	Area of sector $= \frac{\theta}{360} \times \pi r^2$ (1); $\frac{50}{360} \times \frac{22}{7} \times 42^2$ (1); Area $= 770$ cm^2 (1).	1+1+1
30	Volume of cylinder $= \pi \times 9^2 \times 32$ (1); Volume of a sphere $= \frac{4}{3}\pi(6)^3$; $n = \frac{3r^2h}{4\rho^3}$ (1); $n = 9$ (1).	1+1+1
31	$\sum f_i = 100$ and $\sum f_i x_i = 7530$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7530}{100} = 75.3$ (1); $l = 80$, $f_1 = 23$, $f_0 = 20$, $f_2 = 2$, $h = 10$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 80 + \frac{3}{24} \times 10 = 81.25$ (1).	1+1+1
	OR: Total number of outcomes $= 6 \times 6 = 36$ (1); Favourable outcomes: (3,6), (4,5), (5,4), (6,3); number $= 4$ (1); $P = \frac{4}{36} = \frac{1}{9}$ (1).	1+1+1
32	Required number $= \text{HCF}(1250, 9376)$ (1); $\text{HCF}(1250, 9376) = 2$ (2).	1+2
	OR: $(\alpha + \beta)^2 - (\alpha - \beta)^2 = 4\alpha\beta$ (1); $100 - 16 = 4c$ (1); $c = 21$ (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $3n + 2p = 70$ and $2n + 3p = 65$ (1). (ii) Multiply by 3 and 2: $9n + 6p = 210$, $4n + 6p = 130$; subtract: $5n = 80$, so $n = ₹\,16$ (1). (iii) $3(16) + 2p = 70 \Rightarrow p = ₹\,11$ (1). (iv) 6 notebooks cost ₹ 96; remaining ₹ 104; $104 \div 11 = 9$ remainder 5, so 9 pens (1) and ₹ 5 left (1).	1+1+1+2
34	(i) $x(x + 3) = 180 \Rightarrow x^2 + 3x - 180 = 0$ (1). (ii) $(x + 15)(x - 12) = 0$, so $x = 12$ (rejecting -15) (1). (iii) Breadth 12 m, length 15 m (1). (iv) Perimeter $= 2(15 + 12) = 54$ m; wire $= 3 \times 54 = 162$ m (1); cost $= 162 \times 20 = ₹\,3240$ (1).	1+1+1+2
35	Slant height of the cone $l = \sqrt{49^2 + 168^2} = 175$ cm (2); TSA = CSA of cone + CSA of hemisphere $= \pi rl + 2\pi r^2$ (2); TSA $= \frac{22}{7} \times 49 \times 175 + 2 \times \frac{22}{7} \times 49^2 = 42042$ cm^2 (1).	2+2+1
36	Median class is 230–240 with $f = 25$, and the cumulative frequency before it is $cf = 20 + x$ (2); $238.4 = 230 + \frac{50 - (20 + x)}{25} \times 10$ gives $x = 9$ (2); $x = 9$; then $y = 100 - 82 - x = 9$ (1).	2+2+1
	OR: Total outcomes $= 36$ (2); Favourable outcomes: 4; 8; 15 (2); Probabilities: $\frac{1}{9}$, $\frac{2}{9}$, $\frac{5}{12}$ (1).	2+2+1
37	If $1/\sqrt{2} = r$ were rational then $\sqrt{2} = 1/r$ would be rational - contradiction (3); $\frac{3}{\sqrt{2}} = 3 \times \frac{1}{\sqrt{2}}$ is a non-zero rational times an irrational, hence irrational (2).	3+2
	OR: $\alpha + \beta = 3$, $\alpha\beta = -10$ (1+1); new sum $= -7$ (1); new product $= -30$ (1); polynomial $x^2 + 7x - 30$ (1).	1+1+1+1+1
38	Let train u, bus v km/h; $p = \frac{1}{u}$, $q = \frac{1}{v}$: $150p + 210q = \frac{41}{4}$, $240p + 120q = 11$ (2); $p = \frac{1}{30}$, $q = \frac{1}{40}$ (2); train 30 km/h, bus 40 km/h (1).	2+2+1
	OR: Boat speed x: $\frac{48}{x - 2} + \frac{48}{x + 2} = 7$ (1); $48(2x) = 7(x^2 - 4)$ (1); roots of the resulting quadratic: $x = 14$ or negative (1); speed 14 km/h (1).	1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Full Syllabus (Chapters 1–14) | Class 10 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** 269.8	$\sum f_i = 100$ and $\sum f_i x_i = 26980$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{26980}{100} = 269.8$
2	**(d)** 45	$\frac{x}{15+x} = 3 \times \frac{15}{15+x}$ gives $x = 3 \times 15$; $x = 45$
3	**(d)** 3	$12 = 2^2 \times 3$; multiply by the primes that occur to an odd power
4	**(c)** $\frac{5}{2}$	$\dfrac{\alpha^2+\beta^2}{\alpha\beta}$
5	**(b)** 12	Intercepts 6 and 4; area $= \frac{1}{2} \times 6 \times 4$
6	**(a)** 2	Substitute the root
7	**(a)** 4	$d = 2p$
8	**(d)** 14	Sides in ratio $a : b$
9	**(b)** 6	$x = 0$: $mc + na = 0$
10	**(b)** $\frac{1}{3}$	$(\sec + \tan)(\sec - \tan) = 1$
11	**(b)** 12 m	Both objects are on the same side, so the distances are subtracted; Distance $= 12$ m
12	**(b)** 120 cm	Half-chord $= \sqrt{68^2 - 32^2} = 60$ cm; Chord $= 2 \times 60 = 120$ cm
13	**(b)** 117 cm	Arc $= \frac{45}{360} \times 2\pi \times 42 = 33$ cm; Perimeter $= 33 + 84 = 117$ cm
14	**(d)** 14 cm	Cancel $\frac{1}{3}\pi r^2$: $h = 2r$; $h = 14$ cm
15	**(d)** 108	Median class 100–120: $l = 100$, $cf = 44$, $f = 15$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 100 + \frac{50-44}{15} \times 20 = 108$
16	**(d)** 5	Cross-multiplying: $7(7 + x) = 4(16 + x)$; $x = 5$
17	**(a)** Both A and R are true and R is the correct explanation of A.	With h fixed, smaller d gives larger $\tan\theta$, hence larger θ; R is the explanation.
18	**(a)** Both A and R are true and R is the correct explanation of A.	$\angle OPT = 90^\circ$, so $OT^2 = 5^2 + 12^2 = 169$ and $OT = 13$ cm; R is what makes the triangle right-angled.
19	**(a)** Both A and R are true and R is the correct explanation of A.	$\frac{1}{4} \times \frac{22}{7} \times 49 = 38.5$; R is the formula used.
20	**(a)** Both A and R are true and R is the correct explanation of A.	$2\pi r^2 + \pi r^2 = 3\pi r^2$; R gives the reason.

SECTION — B (Very Short Answer) — Answers — 6 × 2 = 12

Q.	Expected Answer / Value Points	Marks
21	$\sum f_i = 83 + p$ and $\sum f_i x_i = 23150 + 250p$; $\frac{23150+250p}{83+p} = 274$ gives $(250 - 274)p = -408$ (1); $p = 17$ (1).	1+1
22	Total number of outcomes $= 6 \times 6 = 36$; Favourable outcomes: (1,3), (2,2), (2,6), (3,1), (3,5), (4,4), (5,3), (6,2), (6,6); number $= 9$ (1); $P = \frac{9}{36} = \frac{1}{4}$ (1).	1+1
23	Each remainder is short of the divisor by 5 (1); LCM$(12, 18) = 36$, number $= 36 - 5 = 31$ (1).	1+1
24	Sum $= 6$, product $= k$ (1); $k = 6$ (1).	1+1
25	Adding: $8(x + y) = 88$ so $x + y = 11$ (1); subtracting: $x - y = 3$, so $x = 7$, $y = 4$ (1).	1+1
26	$\alpha + \beta = 6$, $\alpha\beta = 5$ (1); $\alpha^2 + \beta^2 = 6^2 - 2(5) = 26$ (1).	1+1

SECTION — C (Short Answer) — Answers — 6 × 3 = 18

Q.	Expected Answer / Value Points	Marks
27	Need $\dfrac{a+b}{3} = \dfrac{a-b}{1} = \dfrac{14}{7} = 2$ (1); $a + b = 6$, $a - b = 2$ (1); $a = 4$, $b = 2$ (1).	1+1+1
28	$D = 25 - 4k \geq 0$ (1); $k \leq \frac{25}{4}$ (1); greatest integer $k = 6$ (1).	1+1+1
29	$d = 4$, $a = 6$ (2); $S_{16} = 576$ (1).	2+1
30	Altitude bisects the base: $h^2 = (6)^2 - 3^2 = 27$ (1+1); $h = 3\sqrt{3}$ cm (1).	1+1+1

31	P divides AB in $1:2$: $(3,\ 2)$ (1.5); Q divides AB in $2:1$: $(6,\ 4)$ (1.5).	1.5+1.5
	OR: Square both sides: $1+2\sin\theta\cos\theta=\frac{49}{25}$ (2); $\sin\theta\cos\theta=\frac{12}{25}$ (1).	2+1
32	Distance of the point from the building: $d=14\cot 45^\circ=14$ m (1); Height of the top of the flagstaff $=d\tan 60^\circ=14\sqrt{3}$ m (1); Length $=14\sqrt{3}-14=14\sqrt{3}-14$ m (1).	1+1+1
	OR: $AC=85$ cm; the incircle touches AB, BC at F, D and $BDOF$ is a square of side r (1); $AF=AB-r=36-r$, $CD=BC-r=77-r$ and $AF+CD=AC$ (tangent lengths) (1); $(36-r)+(77-r)=85$ gives $r=14$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers

6 × 5 = 30

Q.	Expected Answer / Value Points	Marks
33	(i) $a=100$, $d=20$ (1). (ii) $a_{10}=100+9\times 20=₹280$ (1). (iii) $S_{12}=\frac{12}{2}[2(100)+11(20)]=6\times 420=₹2520$ (1). (iv) $100+(n-1)20=400\Rightarrow n=16$ (1); $S_{16}=\frac{16}{2}(100+400)=₹4000$ (1).	1+1+1+2
34	(i) AA (each has a right angle and the same sun-angle) (1). (ii) $\frac{h}{24}=\frac{1.5}{2}\Rightarrow h=18$ m (1). (iii) $\frac{6}{s}=\frac{1.5}{2}\Rightarrow s=8$ m (1). (iv) $\sqrt{18^2+24^2}=\sqrt{900}=30$ m by the Pythagoras theorem (2).	1+1+1+2
35	Diagonals bisect each other: midpoint of AC is $(1,2)$ (2); this is also the midpoint of BD (1); $D=(1,\ 4)$ (2).	2+1+2
36	Divide numerator and denominator by $\cos A$ (1): $\frac{5\tan A+4}{7\tan A+3}$ (1); $=\frac{5(\frac{24}{7})+4}{7(\frac{24}{7})+3}=\frac{148}{189}$ (2); verified with $\sin A=\frac{24}{25}$, $\cos A=\frac{7}{25}$ (1).	1+1+2+1
	OR: Let height $=h$; distances $h\cot 60^\circ$ and $h\cot 30^\circ$ add up to 48 (2); $h\left(\frac{1}{\sqrt{3}}+\sqrt{3}\right)=48$ gives $h=12\sqrt{3}$ (2); Height $12\sqrt{3}$ m; distances 12 m and 36 m (1).	2+2+1
37	Area of $\triangle OAP=\frac{1}{2}\times 72\times 21$ and also $=\frac{1}{2}\times OP\times AM$ where $AM\perp OP$ (M is the midpoint of AB) (2); $AM=\frac{72\times 21}{75}=20.16$ cm (2); $AB=2AM=40.32$ cm (1).	2+2+1
	OR: Required area $=\frac{\theta}{360}\times\pi(R^2-r^2)$ (2); $\frac{36}{360}\times\frac{22}{7}\times(21^2-14^2)$ (2); Area $=77$ cm^2 (1).	2+2+1
38	The cylinder has radius r and height $2r$ (2); Cylinder $=\pi r^2(2r)=196465.5$ cm^3; sphere $=\frac{4}{3}\pi r^3=130977$ cm^3 (2); Unoccupied $=65488.5$ cm^3 (1).	2+2+1
	OR: $\sum f_i=100$ and $\sum f_i x_i=7880$; Mean $=\frac{\sum f_i x_i}{\sum f_i}=\frac{7880}{100}=78.8$ (2); $l=50$, $f_1=23$, $f_0=7$, $f_2=19$, $h=10$ (2); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=50+\frac{16}{20}\times 10=58$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER